Linear Regression

Agenda

- Overview of Machine Learning
- Simple Linear Regression
- Multiple Linear Regression
- Handling Categorical Variables
- Evaluation Metrics for Regression

Let's begin the discussion by answering a few questions

Linear Regression Quiz

Which of the following tasks is typically associated with regression problems?

A. Predicting whether an email is spam or not

B. Segmenting customers into different clusters based on their purchasing behavior

C. Predicting the price of a house based on its features

D. Classifying images of handwritten digits into one of ten categories

Linear Regression Quiz

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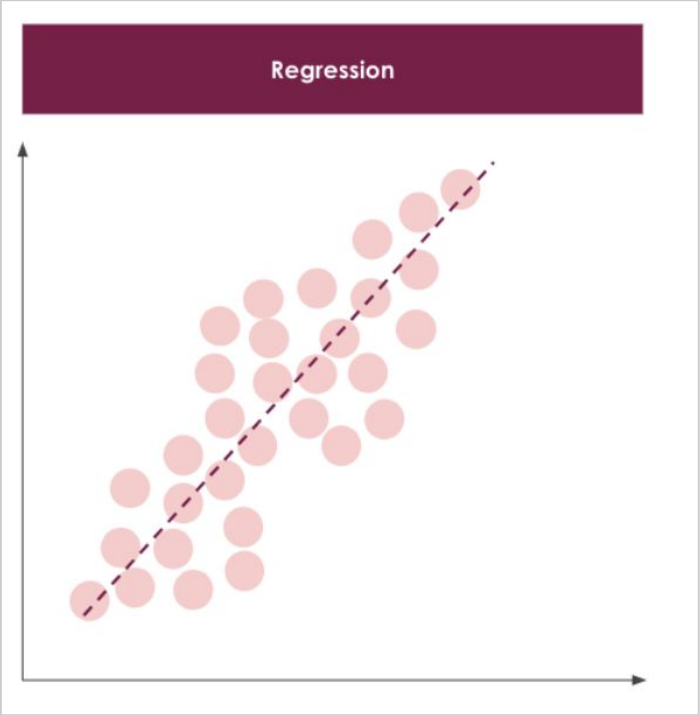
Simple Linear Regression

Regression is a type of **supervised learning task**, which focuses on **predicting a continuous numerical value or quantity** with the given input features.

This can be thought of more as **drawing a "trend line" in vector space**

Predicting the price of a house based on its features (such as size, number of bedrooms, location, etc.), is a classic example of a regression task. The goal is to predict a continuous value (the price) based on various features.



When we did EDA, we plotted our data as scatterplots to spot trends. Now we're adding a regression line on top of those points to turn that visual trend into a model.

Linear Regression Quiz

In a linear regression model,

house_price = 50,000 + 200*area(sq ft) + 150*number_of_rooms

For a 1200 sq ft house with 3 rooms, what will be the price?

A — $ 50,450

B — $ 290,000

C — $ 290,450

D — $ 290,500

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Multiple Linear Regression

To find the value of the **dependent variable** (**house_price**) for the given test data point, we substitute the values of the **independent variables** (**area = 1200 sq ft** and **number_of_rooms = 3**) into the **regression equation**.

house_price = 50,000 + 200 * (area sq ft) + 150 * (number of rooms)

house_price = 50,000 + 200 * 1200 + 150 * 3

house_price = 50,000 + 240,000 + 450

house_price = 290,450

So, the predicted house price for the given test data point is $290,450.

Multiple Linear Regression

Dependent Variable
The variable that is being predicted or explained in the regression model

Independent Variables
The input variables that influence the outcome of the independent variable

$$y = b_0 + b_1 x_1 + b_2 x_2 + \cdots$$

Intercept
The constant term that represents the value of the dependent variable when all independent variables are zero

Coefficients
The weights assigned to each independent variable, indicating their impact on the dependent variable

Linear Regression Quiz

Given a linear regression equation:

$$y = b_0 + b_1x_1 + b_2x_2$$

If x_2 increases by a value of 2, what will be the corresponding increase in the value of y?

A $2b_0$

B $2b_1$

C $2b_2$

D b_2

Linear Regression Quiz

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$$y = b_0 + b_1x_1 + b_2x_2$$

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Multiple Linear Regression Interpretation

From the original equation:

$$y = b_0 + b_1x_1 + b_2x_2$$

We can see that the dependent variable(y) is related to the independent variable x_2 by a factor of b_2.

This means that for every unit change in x_2, there will be a corresponding b_2 change in the dependant variable y.

So, if x_2 increases by **2 units**, y will increase by **$2b_2$ units**.

Linear Regression Quiz

Why can't categorical variables be directly used in linear regression?

A. They have a small variance.

B. They add little value to the prediction power of a model.

C. They represent categories while the best fit line needs to be fit on numerical values.

C. They always cause overfitting, even when properly encoded.

Linear Regression Quiz

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- C They represent categories while the best fit line needs to be fit on numerical values.
- D They always cause overfitting, even when properly encoded.

Encoding of Categorical Variables

Categorical variables are not numbers - even if they might be represented by numbers

Can’t be utilized directly in a linear regression model

Need to be converted into a numerical format

Encoding

Label Encoding

Used when the categories have an inherent sense of order

One-hot Encoding

Used when the categories have no inherent sense of order

Linear Regression Quiz

Which of the following metrics measures the proportion of total variance explained by the model without adjusting for the number of predictors?

A. Mean Absolute Error(MAE)

B. Mean Squared Error (MSE) / Root Mean Squared Error (RMSE)

C. R-Squared (R^2)

D. Adjusted R-Squared (Adj. R^2)

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Evaluation Metrics in Regression

R-squared	Adjusted R-squared	Mean Absolute Error	Root Mean Square Error
• Measure of the % of variance in the target variable explained by the model • Generally the first metric to look at for linear regression model performance • Higher the better	• Conceptually, very similar to R-squared but penalizes for the addition of too many variables • Generally used when you have too many variables as adding more variables always increases R^2 but not Adjusted R^2 • Higher the better	• Simplest metric to check prediction accuracy • Same unit as the dependent variable • Not sensitive to outliers i.e. errors doesn’t increase too much if there are outliers • Difficult to optimize from a mathematical point of view (pure maths logic) • Lower the better	• Another metric to measure the accuracy of prediction • Same unit as the dependent variable • Sensitive to outliers - errors will be magnified due to the square function • But has other mathematical advantages that will be covered later • Lower the better

Power Ahead !

Appendix

Metric	Formula
R-squared	$R^2 = 1 - \frac{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2}$
Adjusted R-squared	$Adj.R^2 = 1 - \left[\frac{(1 - R^2)(n - 1)}{n - k - 1}\right]$
Mean Absolute Error	$MAE = \frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i \rvert$
Root Mean Square Error	$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$